

OpenAI Cost Breakdown — WA Images (Jewelry Finder)

Models in Use

Purpose	Model
Image profiling & query analysis	gpt-4.1
Semantic search embeddings	text-embedding-3-large

Current OpenAI Pricing

Model	Input	Output
gpt-4.1	\$2.00 / 1M tokens	\$8.00 / 1M tokens
text-embedding-3-large	\$0.13 / 1M tokens	— (no output tokens)

Cost Per Operation

1. Customer sends an IMAGE query

Triggers `analyze_image_profile` (gpt-4.1, `detail: "high"`) + `embed_text` .

Component	Tokens	Cost
PROFILE_PROMPT (system)	~900 input	—
Image at high detail (typical jewelry photo)	~765–1,100 input	—
JSON profile output	~400 output	—
Total per image query	~1,700 in / 400 out	~\$0.0065
<code>embed_text</code>	~350 input	~\$0.00005
Total		≈ \$0.007

2. Customer sends a TEXT query

Triggers `analyze_query` (gpt-4.1, text only, max 30 output tokens) + `embed_text` .

Component	Tokens	Cost
SYSTEM_PROMPT + user text	~370 input	—
Classification output	~10 output	—
<code>embed_text</code>	~350 input	~\$0.00005
Total per text query		≈ \$0.0009

3. Indexing a new image (one-time per image)

Same pipeline as an image query — `analyze_image_profile` + `embed_text` .

	Cost
Per image indexed	≈ \$0.007

Monthly Cost Estimates

Scale	Catalog Size	Queries / Day	One-Time Indexing	Monthly Recurring
Small	500 images	30 image + 50 text	~\$3.50	~\$8-10
Medium	2,000 images	100 image + 100 text	~\$14	~\$25-30
Large	5,000 images	300 image + 200 text	~\$35	~\$70-80

The dominant cost is **image queries** (~\$0.007 each). Text queries are ~8× cheaper.